

ϕ TD-Gaussian

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.06193	0.1578	0.1349	0.08572
Cramér $\ell_2^2 \downarrow$	0.0006799	0.002792	0.003143	0.001509
CF - $\ell_2^2/\omega^2 \downarrow$	0.0005816	0.001935	0.00271	0.001245

ϕ TD-Cauchy

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1023	0.04463	0.07612	0.1336
Cramér $\ell_2^2 \downarrow$	0.002676	0.0003078	0.001698	0.002573
CF - $\ell_2^2/\omega^2 \downarrow$	0.001773	0.0002583	0.001001	0.001604

ϕ TD-Gamma

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.7105	2.582	1.947	0.1248
Cramér $\ell_2^2 \downarrow$	0.2982	0.811	0.7369	0.002237
CF - $\ell_2^2/\omega^2 \downarrow$	0.2625	0.7452	0.6962	0.002369

ϕ TD-Categorical

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.2841	0.3802	0.3289	0.2999
Cramér $\ell_2^2 \downarrow$	0.0281	0.01812	0.03693	0.02875
CF - $\ell_2^2/\omega^2 \downarrow$	0.02112	0.01423	0.03047	0.02315

ϕ TD-Quantile

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1022	0.224	0.1357	0.1796
Cramér $\ell_2^2 \downarrow$	0.002186	0.003316	0.003922	0.003717
CF - $\ell_2^2/\omega^2 \downarrow$	0.001651	0.003736	0.003317	0.0033